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Real Analysis
Assignment - 2

Question 1:

(a) Let, $(x,y) \in \left(\bigcup_{\alpha\in I} A_\alpha\right) \times \left(\bigcup_{\beta\in J} B_\beta\right)$

$\Rightarrow x \in \bigcup_{\alpha\in I} A_\alpha$ and $y \in \bigcup_{\beta\in J} B_\beta$ [definition]

$\Rightarrow x \in A_\alpha$ for $\alpha \in I$ and $y \in B_\beta$ for $\beta \in J$ [definition]

$\Rightarrow (x,y) \in A_\alpha \times B_\beta$ for $\alpha \in I, \beta \in J$ [definition]

$\Rightarrow (x,y) \in A_\alpha \times B_\beta$ for $(\alpha,\beta) \in (I \times J)$ [definition]

$\Rightarrow (x,y) \in \bigcup_{(\alpha,\beta)\in(I\times J)} (A_\alpha \times B_\beta)$ [definition]

$\therefore \left(\bigcup_{\alpha\in I} A_\alpha\right) \times \left(\bigcup_{\beta\in J} B_\beta\right) \subseteq \bigcup_{(\alpha,\beta)\in(I\times J)} (A_\alpha \times B_\beta)$

Let, $(x,y) \in \bigcup_{(\alpha,\beta)\in I\times J} (A_\alpha \times B_\beta)$

$\Rightarrow (x,y) \in A_\alpha \times B_\beta$ for $(\alpha,\beta) \in I \times J$ [definition]

$\Rightarrow x \in A_\alpha$ for $\alpha \in I$ and $y \in B_\beta$ for $\beta \in J$ [2 definitions for compactness]

$\Rightarrow x \in \bigcup_{\alpha\in I} A_\alpha$ and $y \in \bigcup_{\beta\in J} B_\beta$ [definition]

$\Rightarrow (x,y) \in \left(\bigcup_{\alpha\in I} A_\alpha\right) \times \left(\bigcup_{\beta\in J} B_\beta\right)$ [definition]

$$\therefore \bigcup_{(\alpha,\beta)\in I\times J} (A_\alpha \times B_\beta) \subseteq \left(\bigcup_{\alpha\in I} A_\alpha\right) \times \left(\bigcup_{\beta\in J} B_\beta\right)$$

$$\therefore \bigcup_{(\alpha,\beta)\in I\times J} (A_\alpha \times B_\beta) = \left(\bigcup_{\alpha\in I} A_\alpha\right) \times \left(\bigcup_{\beta\in J} B_\beta\right)$$

(b) In the previous result, let,

$I = \{1,2\} = J$

$A_1 = A$, $A_2 = C$, $B_1 = B$ and $B_2 = D$

Then, LHS $= \bigcup_{(\alpha,\beta)\in \{1,2\}\times\{1,2\}} (A_\alpha \times B_\beta)$

$= \bigcup_{(\alpha,\beta)\in \{(1,1),(1,2),(2,1),(2,2)\}} (A_\alpha \times B_\beta)$

$= (A_1 \times B_1) \cup (A_1 \times B_2) \cup (A_2 \times B_1) \cup (A_2 \times B_2)$

$= (A \times B) \cup (A \times D) \cup (C \times B) \cup (C \times D)$

And, RHS $= \left(\bigcup_{\alpha\in\{1,2\}} A_\alpha\right) \times \left(\bigcup_{\beta\in\{1,2\}} B_\beta\right)$

$= (A_1 \cup A_2) \times (B_1 \cup B_2)$

$= (A \cup C) \times (B \cup D)$

$\therefore (A \cup C) \times (B \cup D) = (A \times B) \cup (A \times D) \cup (C \times B) \cup (C \times D)$

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(c) Let, $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$

$= (A \times B) \cup (A \times D) \cup (C \times B) \cup (C \times D)$ — ② [previous result]

Let, $(A \times B) \cup (C \times D) = E$ [for abbreviation] — ①

We know, $(A \times B) \subseteq E$ and $C \times D \subseteq E$. [from ①]

But for ② to be true, $A \times D \subseteq E$ and $C \times B \subseteq E$ should also hold.

For $A \times D \subseteq E$:

Proving $A \times D \subseteq E \Rightarrow (A \subseteq C) \vee (D \subseteq B)$:

~~If $A \times D \subseteq E$,~~

Case I: $D \subseteq B$

We are already done since $(A \times B \subseteq E) \wedge (D \subseteq B) \rightarrow (A \times D \subseteq E)$.

[This is a standard result].

Case II: $D \not\subseteq B$

$\Rightarrow \exists\, d \in D,\ d \notin B$

Let an $a \in A$.

Then, $(a, d) \in A \times D$

$\Rightarrow (a, d) \in E$ — ③ [hypothesis]

But $(a, d) \in A \times B$ — ④

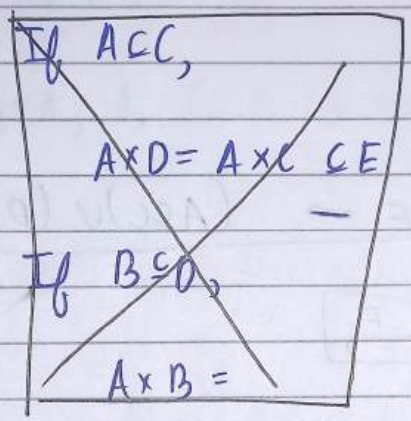
③ and ④ imply $(a,d) \in (C \times D)$

$\Rightarrow a \in C$

$\therefore A \subseteq C$

$\therefore A \times D \subseteq E \rightarrow (B \subseteq D) \vee (A \subseteq C)$

Proving $(A \subseteq C) \vee (B \subseteq B) \rightarrow A \times D \subseteq E$:



If $A \subseteq C$,

$A \times D \subseteq E$ because $C \times D \subseteq E$. [standard result]

If $D \subseteq B$,

$A \times D \subseteq E$ because $A \times B \subseteq E$. [standard result]

$\therefore$ it holds

Thus, $(A \times D) \subseteq E \leftrightarrow (A \subseteq C) \vee (D \subseteq B)$.

We can similarly prove, $C \times B \subseteq E \leftrightarrow (C \subseteq A) \vee (D \subseteq B)$.

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Therefore, $E = (A \cup C) \times (B \cup D)$

$\Leftrightarrow E = (A \times B) \cup (A \times D) \cup (C \times B) \cup (C \times D)$

$\Leftrightarrow (A \times D) \subseteq E$ and $(C \times B) \subseteq E$

$\Leftrightarrow ((A \subseteq C) \vee (D \subseteq B)) \wedge ((C \subseteq A) \vee (D \subseteq B))$

Therefore proved.

(d) Proving $(A \setminus C) \times (B \setminus D) \subseteq (A \times B) \setminus (C \times D)$,

Let, $(a, b) \in (A \setminus C) \times (B \setminus D)$

$\Rightarrow a \in A, b \in B, a \notin C, b \notin D$

$\Rightarrow (a, b) \in A \times B, (a, b) \notin (C \times D)$

$\Rightarrow (a, b) \in (A \times B) / (C \times D)$

Hence, $(A \setminus C) \times (B \setminus D) \subseteq (A \times B) \setminus (C \times D)$.

This inclusion is automatic.

Let, equality holds.

We want to show $A \cap C = \phi$ or $B \subseteq D$.

Let, $A \cap C \neq \phi$

Let, $a \in A \cap C$

Let, $b \in B$.

If $b \notin D$, $a \in A$ and $b \in B$

$(a, b) \in A \times B$

But, $a \in C$, $b \notin D$.

$\Rightarrow (a, b) \notin (C \times D)$

$\Rightarrow (a, b) \in (A \times B) \setminus (C \times D)$.

~~By assumed equality, $(a,b) \in (A \setminus C) \times (B \setminus D)$~~
$\Rightarrow b \in B \setminus D$

Contradicting $b \in D$

$\therefore a \in C$
$\therefore A \subseteq C$

Hence either $A \subseteq C$ or $B \cap D = \phi$

Similarly, we can show

By assumed equality, $(a,b) \in (A \setminus C) \times (B \setminus D)$
$\Rightarrow a \in A \setminus C$

Contradicting $a \in C$
$\therefore b \in D$
$\therefore B \subseteq D$

$\therefore A \cap C = \phi$ or $B \subseteq D$

Similarly, we can show either $A \subseteq C$ or $B \cap D = \phi$.

We can show that $(A \cap C = \phi \vee B \subseteq D) \wedge (A \subseteq C \vee B \cap D = \phi)$ imply the equality and second inclusion.

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Problem 2:

(a) $f: X \to Y$
$g: Z \to W$

$f \times g : X \times Z \to Y \times W \quad (f \times g)(x, z) := (f(x), g(z))$

For $f \times g$ to be a function, every element of $X \times Z$ must be mapped to exactly one element of $Y \times Z$.

Let any, $(x, z) \in X \times Z$

$\Rightarrow x \in X$ and $z \in Z$

Since $f: X \to Y$, $f(x) \in Y$ and $f(x)$ is unique
Since $g: Z \to W$, $g(z) \in W$ and $g(z)$ is unique.

$f \times g$ maps $(x, z) \mapsto (f(x), g(z)) \in Y \times W$.

And, Since $f(x)$ and $g(z)$ are uniquely determined by x and z, $(f(x), g(z))$ is uniquely determined by (x, z).

$\therefore f \times g$ is a function

Now, let $(y, w) \in (f \times g)(A \times B)$

By definition, $\exists (x, z) \in A \times B$ st $(f \times g)(x, z) = (y, w)$.
$= (f(x), g(z))$

Also, $x \in A$ and $z \in B$
And, $y = f(x)$ and $w = g(z)$

Therfore, $y = f(x) \in f(A)$
$w = g(z) \in g(B)$

$\therefore (y, w) \in f(A) \times g(B)$

$\therefore (f \times g)(A \times B) \subseteq f(A) \times f(B)$.

Now, let $(y, w) \in f(A) \times g(B)$

$\Rightarrow y \in f(A)$ and $w \in g(B)$

$\Rightarrow \exists\ x \in A$ st $f(x) = y$ and
$\exists\ z \in B$ st $f(z) = w$

Also, $(x, y) \in A \times B$

Then, $(f \times g)(x, y) = (f(x), g(y)) = (y, w)$

Therfore, $(y, w) \in (f \times g)(A \times B)$

$\therefore f(A) \times f(B) \subseteq (f \times g)(A \times B)$

Therfore, $f(A) \times f(B) = (f \times g)(A \times B)$ $\quad A \subseteq X,\ B \subseteq Y$

(b) Let, $(x, z) \in X \times Z$,

Then, $(x, z) \in (f \times g)^{-1}(U \times V) \Leftrightarrow (f \times g)(x, z) \in U \times V$
$\Leftrightarrow (f(x), g(z)) \in U \times V$
$\Leftrightarrow f(x) \in U \wedge g(z) \in V$
$\Leftrightarrow x \in f^{-1}(U) \wedge z \in g^{-1}(V)$
$\Leftrightarrow (x, z) \in f^{-1}(U) \times g^{-1}(V)$

Therefore, $(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V)$.

Question 3:

$(A^B)^C$ is set of all functions $f: C \to A^B$. For all $c \in C$, $f(c): B \to A$.

$A^{B \times C}$ is set of all functions $g: B \times C \to A$.

Define, $m: (A^B)^C \to A^{B \times C}$

$m(f) := g$

~~We can treat f as a two variable function by defining,~~

~~$f(b,c) := (f(c))(b)$~~

We can define g as, $g(b,c) := (f(c))(b)$

Injectivity:

Let, $m(f_1) = m(f_2)$

$\Rightarrow (f_1(c))(b) = (f_2(c))(b) \quad \forall\, b \in B,\ c \in C$

~~For a specific c,~~

For an arbitrary $c \in C$, $f_1(c)$ and $f_2(c)$ agree $\forall\, b \in B$.

$\therefore f_1 = f_2$

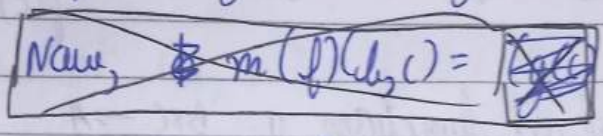
$\therefore$ m is injective.

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Surjectivity:

Let, any $g \in A^{B \times C}$

We can define f by specifying for all $c \in C$, $f(c)$ is given by $b \mapsto g(b, c)$.



Now, ~~$m(f)(b, c) =$~~

$\therefore m(f) = g$ [f can always be constructed]

$\therefore m$ is surjective.

Therefore, m is bijective.

Therefore, $|(A^B)^C| = |A^{B \times C}|$

Now, let, $|A| = a$, $|B| = b$, $|C| = c$ for $a, b, c \in R$

We know, $|(A^B)^C| = |A^B|^{|C|} = (|A|^{|B|})^{|C|} = (a^b)^c$

And, $|A^{B \times C}| = |A|^{|B \times C|} = |A|^{|B||C|} = a^{bc}$

Therefore, $(a^b)^c = a^{bc}$ $a, b, c \in R$

Question 4:

Using notation $\left|\bigcup_{i=1}^{n} A_i\right| = \left|\bigcup_{i \in \{1,\ldots,n\}} A_i\right|$

Suppose for sake of contradiction, no such i exists.

Then, $|A_i| < 2 \Rightarrow |A_i| \leq 1$ $\quad \forall i \in \{1,\ldots,n\}$

~~$\Rightarrow |A_i| = 0$ or $|A_i| = 1$~~

We know, $|X \cup Y| \leq |X| + |Y|$ for any set X, Y.
We can naturally extend this to the union.

$$\left|\bigcup_{i=1}^{n} A_i\right| \leq \sum_{i=1}^{n} |A_i|$$

$$\left|\bigcup_{i=1}^{n} A_i\right| \leq \sum_{i=1}^{n} 1$$

$$\left|\bigcup_{i=1}^{n} A_i\right| \leq n$$

But this contradicts the fact that $\left|\bigcup_{i=1}^{n} A_i\right| > n$.

Therefore, our assumption is false.
Therefore, $\exists i \in \{1,\ldots,n\}$ st $|A_i| \geq 2$.

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Question 5:

Choose $P(n)$: $n \geq 0$

Base case: $P(0)$: $0 \geq 0$

Is true trivially

Inductive step:

Let, $P(n)$ is true.

To prove: $P(n++)$: $n++ \geq 0$

$P(n)$: $n \geq 0$

$\Rightarrow n++ = n+1 \geq 1 > 0$

$\Rightarrow n++ > 0$

$\Rightarrow n++ \geq 0$

$\therefore P(n) \Rightarrow P(n++)$

We can see $P(0)$ is true and $P(n)$ implies $P(n++)$ for all n.

But $P(n)$ isn't true for all integers n. It fails for all negative integers.

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Question 6:

(1) ignore is a rule from $\mathbb{Z}$ to $\mathbb{N}$.

ignore $(a-b) := a$

W1: It does produce a element of $\mathbb{N}$ since both $a, b \in \mathbb{N}$.

W2: We need $a-b = c-d \Rightarrow$ ignore $(a-b)$ = ignore $(c-d)$

That is, $a-b = c-d \Rightarrow a = c$

But that isn't ~~always~~ true.

Ex: $1-0 = 2-1$ because $1+1 = 2 = 0+2$

But ignore $(1-0) = 1 \neq 2 =$ ignore $(2-1)$

∴ ignore isn't well-defined.

(2) mediant is a rule from $\mathbb{Q} \times \mathbb{Q}$ to $\mathbb{Q}$.

~~W1: It does always produce an element of $\mathbb{Q}$ since~~

~~$a+c \in \mathbb{Z}$ and $b+d \in \mathbb{Z}$ and $b+d \neq 0$ because~~

~~W2:~~

W1: We need $(a+c)//(b+d)$ to be a valid element of $\mathbb{Q}$.

Requirements: $a, b, c, d \in \mathbb{Z}$ and $b \neq 0$, $d \neq 0$.

If we take $b \neq 0$ and $d = -b$, all requirements are satisfied.

But, $b+d = b+(-b)=0$

And, $(a+c)//0$ isn't an elmt of Ω.

$\therefore$ W1 already fails.

$\therefore$ mediant isn't well-defined.

Question 7:

(a) Proving (a_n) is rigid $\Rightarrow$ eventual constancy,

$\exists N \geq 1$ st $\forall \epsilon > 0 \quad |a_j - a_k| \leq \epsilon \quad \forall j, k \geq N$

Only non-negative real numbers less than or equal to ϵ, $\forall \epsilon > 0$ is 0.

Choose $\epsilon = 1/2$.

Then, $|a_j - a_k| \leq \frac{1}{2} \quad \forall j, k \geq N$

But, $a_j - a_k \in \mathbb{Z}$

Only integers whose absolute value is less than or equal to $\frac{1}{2}$ is 0.

$\therefore |a_j - a_k| = 0 \quad \forall j, k \geq N$

$a_j = a_k \quad \forall j, k \geq N$

Taking $k = N$, $a_j = a_N \quad \forall j \geq N$

That is, (a_n) is eventually constant.

Proving eventually constancy $\Rightarrow (a_n)$ is rigid.

If its eventually rigid,

$\exists N \geq 1,\ a_j = C \quad \forall j \geq N$

Now, $a_j = a_k = C \quad \forall j, k \geq N$

$\Rightarrow |a_j - a_k| = 0 \quad \forall j, k \geq N$

$\Rightarrow |a_j - a_k| \leq \epsilon \quad \forall j, k \geq N \quad \forall \epsilon > 0$ ~~$\forall \epsilon \geq 0$, $0 < \epsilon$~~

$\therefore (a_n)$ is rigid.

Therefore, (a_n) is rigid iff it is eventually constant.

(b) Definition of Cauchy sequence -

$$\forall \epsilon > 0,\ \exists N,\ |a_j - a_k| \leq \epsilon \quad \forall j, k \geq N$$

In words - A sequence is Cauchy if it is eventually ~~after~~ ϵ- steady $\forall \epsilon > 0$.

Definition of a rigid sequence - A sequence which is ϵ-steady $\forall \epsilon > 0$ after an index N.

We can see that a rigid sequence is eventually ϵ-steady $\forall \epsilon > 0$ [in fact, it becomes ϵ-steady $\forall \epsilon > 0$ at once]

$\therefore$ every rigid sequence is a Cauchy sequence.

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But every ~~non~~ Cauchy sequence isn't rigid.
We can take example of a sequence which is Cauchy but never constant:

$$a_n = \frac{1}{n}$$

Question 8:

(a) (a_n) is Cauchy:

$\forall \epsilon > 0, \exists N, \quad |a_j - a_k| \leq \epsilon \quad \forall j, k \geq N$

Now, $\phi(j) \geq j \geq N$
$\phi(k) \geq k \geq N$

Let, another $m, n \geq N \quad \Rightarrow |a_m - a_n| \leq \epsilon$

Let, $m = \phi(j)$
$n = \phi(k)$

$\therefore |a_{\phi(j)} - a_{\phi(k)}| \leq \epsilon$

$\therefore (a_{\phi(n)})$ is Cauchy

Showing $a_{\phi(n)} \sim a_n$:

Take an $n \geq N$

Then, $\phi(n) \geq n \geq N$

We can apply Cauchy property with

$j = n$ and $k = \phi(n)$

Therefore, $|a_n - a_{\phi(n)}| \leq \epsilon \quad \forall n \geq N, \forall \epsilon > 0$

Therefore, $a_{\phi(n)} \sim a_n$

Both results are proved.

(b) For (a_{n+1}):

Take, $\phi(n) = n+1$
Clearly, $\phi(n) \geq n$

$\therefore$ By previous result, (a_{n+1}) is Cauchy and $(a_{n+1}) \sim a_n$

For (a_{2n}) and (a_{n^2}) we can take $\phi(n) = 2n$ and $\phi(n) = n^2$ respectively. In both cases, $\phi(n) \geq n$.

$\therefore$ Both (a_{2n}) and (a_{n^2}) are also Cauchy and equivalent to (a_n).

(c) Let us drop the ϕ hypothesis $\phi(n) \geq n$.

Define, $a_1 := 0$
$a_n := 1 \quad n \geq 2$

Also define $\phi(n) = 1$.

Then, the sequence $(a_n) = (0, 1, 1, 1, \ldots)$ is Cauchy and equivalent to 1.
While, the sequ $(a_{\phi(n)}) = (a_1, a_1, \ldots) = (0, 0, 0, \ldots)$ is Cauchy but equivalent to 0.

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On more formally,

$$|a_n - a_{\phi(n)}| = |1-0| = 1 \qquad \forall\, n \geq 2$$

Therefore, $a_n \sim a_{\phi(n)}$ doesn't hold if we drop the hypothesis that $\phi(n) \geq n$.